LLM API Integration

OpenAI

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#AI-Dev

Curious about the Final Exam?



- Check the **Regular Exam** section to find:
  - Full Project Description
  - All Requirements
  - Final Upload Deadline:
    - **must submit no later than 16:00 h, 15 December 2026**

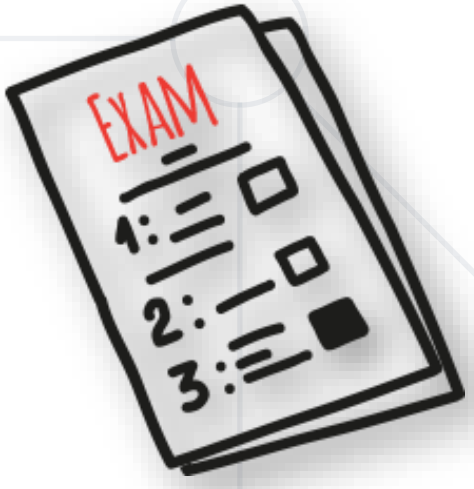
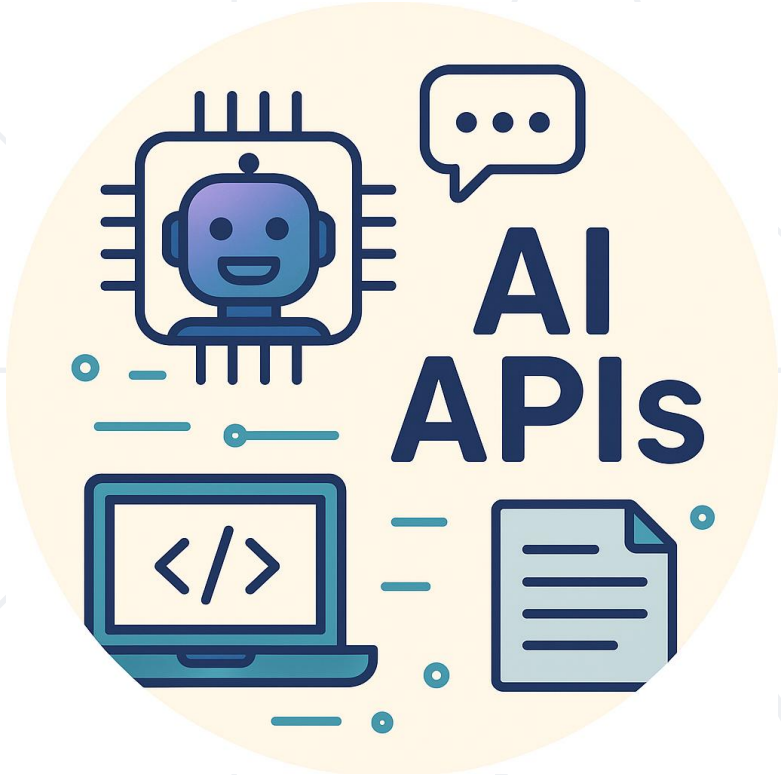


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Goal & Use Cases

Why Use OpenAI and Anthropic APIs?

Why Use OpenAI and Anthropic APIs?

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- Leverage state-of-the-art **language models** in your **own applications**
- Build custom **AI-powered features** beyond the ChatGPT and Claude UI
- **Integrate** language understanding into tools, applications, or workflows

Prompt-Based Interaction

- Send a **prompt** → Get a **response**
- Just like ChatGPT and Claude – but in **code**
- Works with **plain text**, **structured input**, **functions**

```
from openai import OpenAI
client = OpenAI()

response = client.responses.create(
    model="gpt-4.1",
    input="Write a one-sentence bedtime story about a unicorn."
)

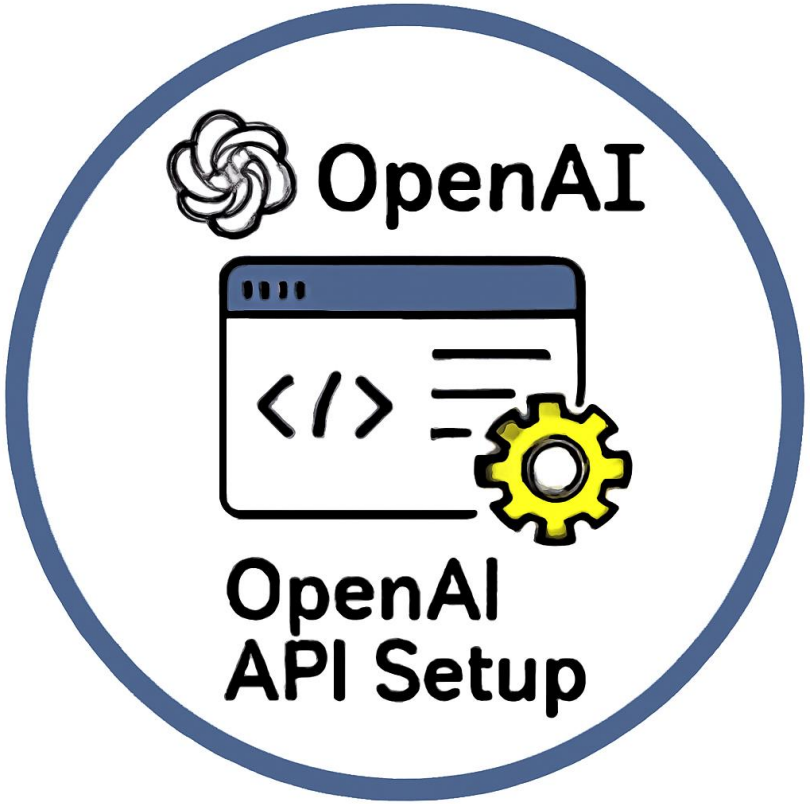
print(response.output_text)
```

Real-World Use Cases

- Generate **emails**, **summaries**, **answers**
- Build smart **chatbots** and **assistants**
- Create **content** or **validate** input
- **Automate** repetitive tasks

Today's Focus

- Text **prompt** & parameters
- **JSON** mode in OpenAI
- Working with **files** in Anthropic
- Dig into **OpenRouter**

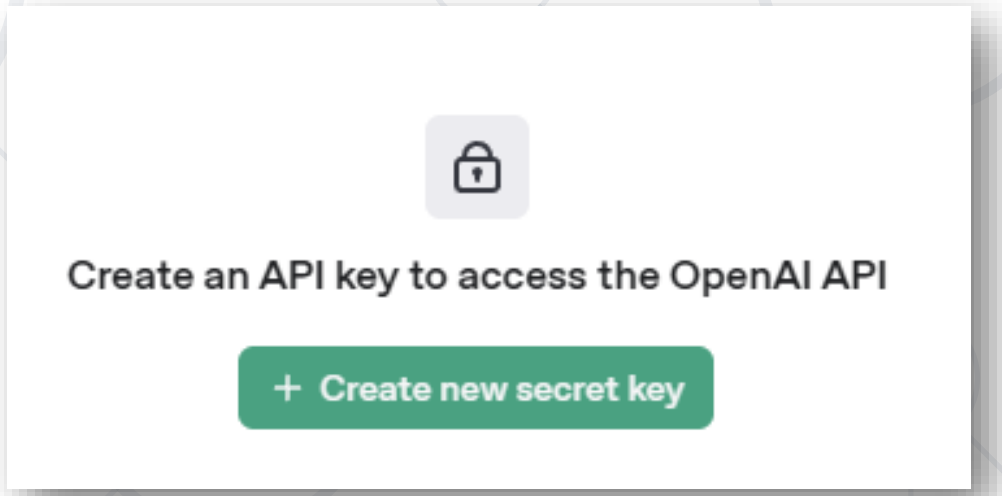


OpenAI API Setup

How to Start Using OpenAI APIs?

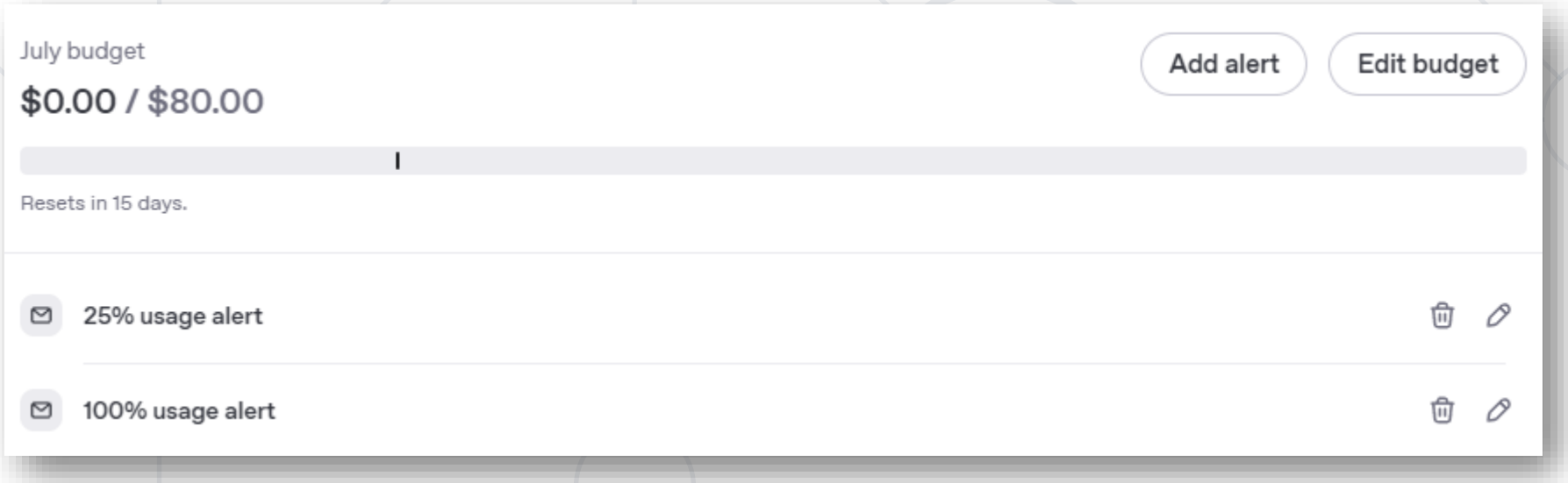
Account Setup

- Create **account** at platform.openai.com
- Get your API **key**



Billing & Pricing

- Pay-as-you-go
- Free tier with usage **limits**
- Depends on amount of **tokens** and **model**
- Can track **expenses** and set **limits**



Rate Limits & Tokens

- **Tokens** = words or parts of words
- Each model has token **limits**
- Monitor usage in **dashboard**

Rate limits ensure fair and reliable access to the API by placing specific caps on requests or tokens used within a given time period. Your usage tier determines how high these limits are set and automatically increases as you send more requests and spend more on the API.

TIER	RPM	TPM	BATCH QUEUE LIMIT
Free		Not supported	
Tier 1	1,000	100,000	1,000,000
Tier 2	2,000	2,000,000	2,000,000

Choosing a Model

- Consider context window, speed, cost
- Advice: start with lowest cost and update if necessary

Text tokens

Price per 1M tokens · Batch API price

Model	Input	Cached input	Output
gpt-4.1 ↪ gpt-4.1-2025-04-14	$2.00	$0.50	$8.00
gpt-4.1-mini ↪ gpt-4.1-mini-2025-04-14	$0.40	$0.10	$1.60
gpt-4.1-nano ↪ gpt-4.1-nano-2025-04-14	$0.10	$0.025	$0.40



OpenAI API Integration

Start Using OpenAI APIs

Basic Request Structure

- Endpoint: *https://api.openai.com/v1/responses*
- Method: POST
- Body: **model**, **input**, max_output_tokens, etc.
- *https://platform.openai.com/docs/api-reference/responses/create*

Official SDKs and Community Libraries

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- Official **SDKs** (recommended): JavaScript, Python, .NET, Java, Go
- *https://platform.openai.com/docs/libraries#community-libraries*

Install the OpenAI SDK with pip

```
pip install openai
```

Test a basic API request

```python
from openai import OpenAI
client = OpenAI()

response = client.responses.create(
    model="gpt-4.1",
    input="Write a one-sentence bedtime story about a unicorn."
)

print(response.output_text)
```

Prompt Structure and Tuning

- **instructions** - a system message inserted into the model's context
- max_output_tokens - limit response length
- previous_response_id - multi-turn conversations

Generate text with instructions

python

```python
from openai import OpenAI
client = OpenAI()

response = client.responses.create(
    model="gpt-4.1",
    instructions="Talk like a pirate.",
    input="Are semicolons optional in JavaScript?",
)

print(response.output_text)
```

Working with JSON

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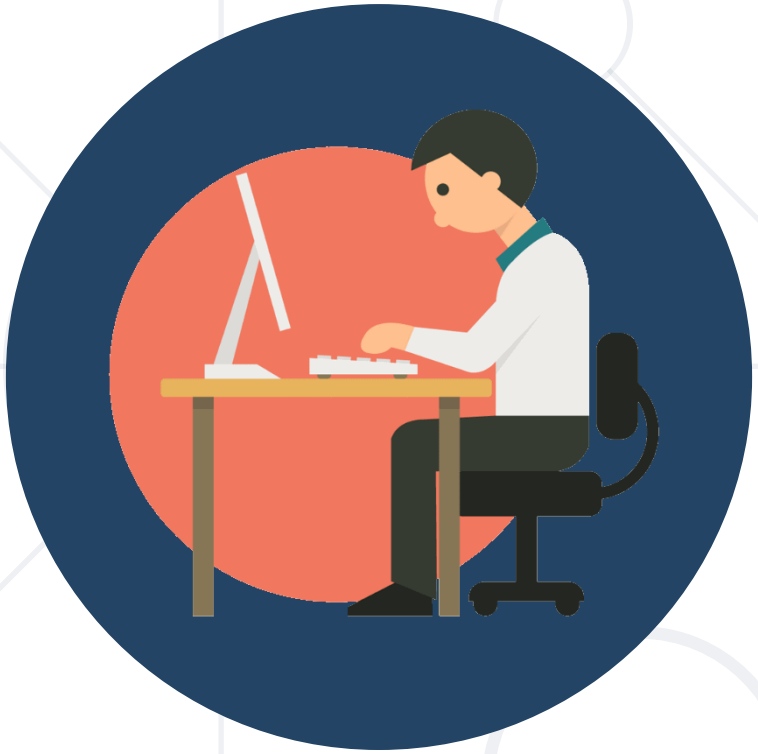
- Enable **JSON** mode
- Get **structured**, parseable output
- Useful for **data extraction**, forms, actions

```python
from openai import OpenAI
from pydantic import BaseModel

client = OpenAI()

class CalendarEvent(BaseModel):
    name: str
    date: str
    participants: list[str]

response = client.responses.parse(
    model="gpt-4o-2024-08-06",
    input=[
        {"role": "system", "content": "Extract the event information."},
        {
            "role": "user",
            "content": "Alice and Bob are going to a science fair on Friday.",
        },
    ],
    text_format=CalendarEvent,
)
```



Practice

Live Exercise – Setup OpenAI API Integration

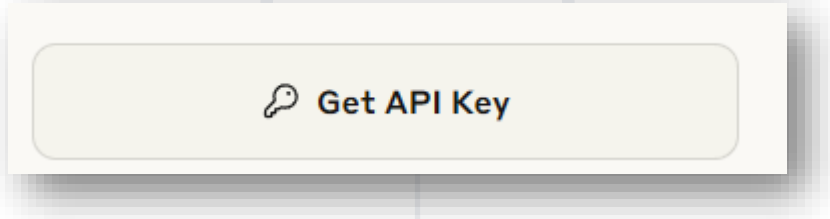


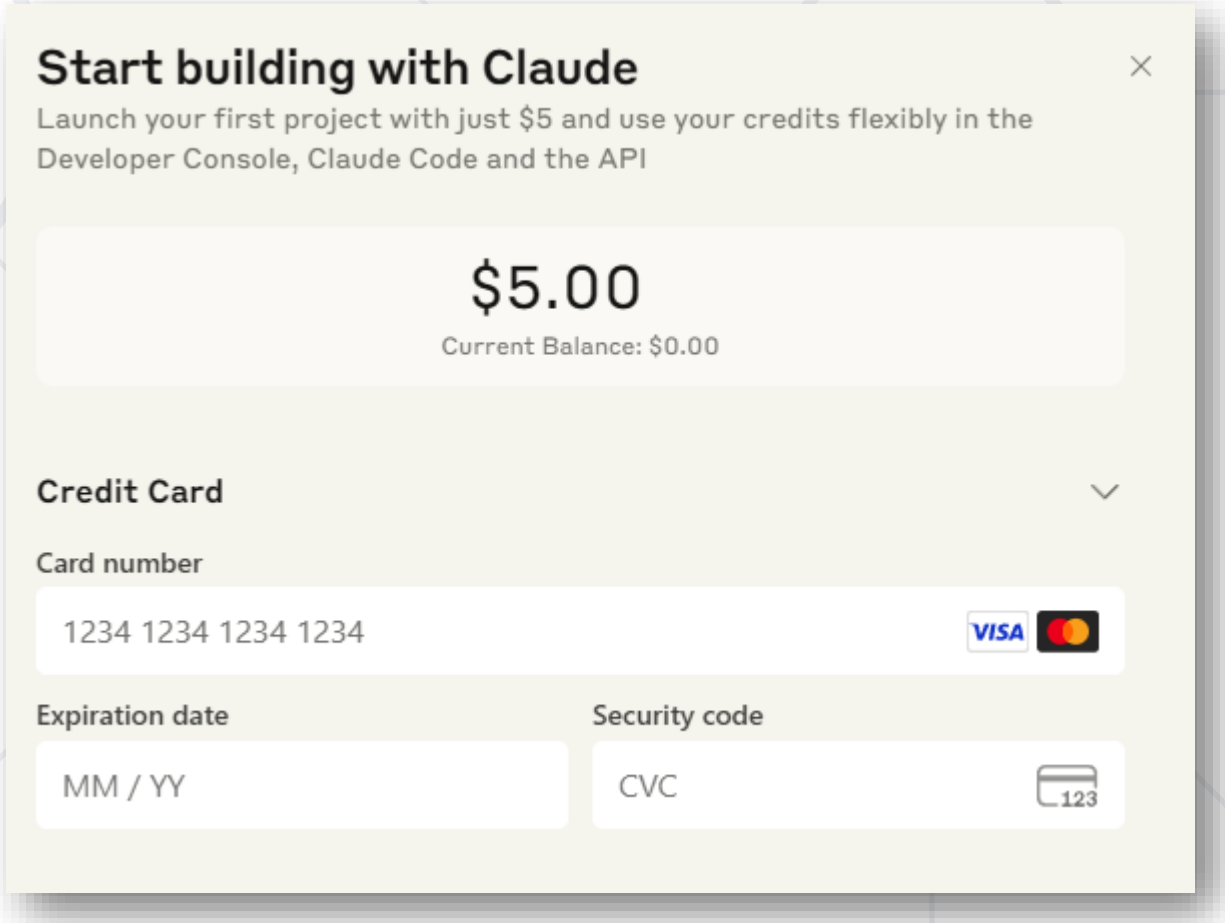
Anthropic API Setup

How to Start Using Anthropic APIs?

Account Setup

- Create **account** at *https://console.anthropic.com/*
- Get your API **key**
- Requires **payment**, no free tier





Billing & Pricing

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- Pay-as-you-go
- No free tier
- Depends on amount of **tokens** and **model**
- Can track **expenses** and set **limits**

Model	Base Input Tokens	5m Cache Writes	1h Cache Writes	Cache Hits & Refreshes	Output Tokens
Claude Opus 4.6	$5 / MTok	$6.25 / MTok	$10 / MTok	$0.50 / MTok	$25 / MTok
Claude Opus 4.5	$5 / MTok	$6.25 / MTok	$10 / MTok	$0.50 / MTok	$25 / MTok
Claude Opus 4.1	$15 / MTok	$18.75 / MTok	$30 / MTok	$1.50 / MTok	$75 / MTok
Claude Opus 4	$15 / MTok	$18.75 / MTok	$30 / MTok	$1.50 / MTok	$75 / MTok
Claude Sonnet 4.6	$3 / MTok	$3.75 / MTok	$6 / MTok	$0.30 / MTok	$15 / MTok
Claude Sonnet 4.5	$3 / MTok	$3.75 / MTok	$6 / MTok	$0.30 / MTok	$15 / MTok
Claude Sonnet 4	$3 / MTok	$3.75 / MTok	$6 / MTok	$0.30 / MTok	$15 / MTok

Rate Limits & Tokens

- **Tokens** = words or parts of words
- **Rate limits** vary by usage tier and affect how many requests you can make

- **Tier 1:** Entry-level usage with basic limits
- **Tier 2:** Increased limits for growing applications
- **Tier 3:** Higher limits for established applications
- **Tier 4:** Maximum standard limits
- **Enterprise:** Custom limits available

Choosing a Model

- Consider context window, speed, cost
- Advice: start with lowest cost and update if necessary

Feature	Claude Opus 4.6	Claude Sonnet 4.6	Claude Haiku 4.5
Description	The most intelligent model for building agents and coding	The best combination of speed and intelligence	The fastest model with near-frontier intelligence
Claude API ID	`claude-opus-4-6`	`claude-sonnet-4-6`	`claude-haiku-4-5-20251001`



Anthropic API Integration

Start Using Anthropic APIs

Basic Request Structure

- Endpoint: *https://api.anthropic.com/v1/messages*
- Method: POST
- Body: **model**, **messages**, max_tokens, etc.
- *https://docs.anthropic.com/en/api/messages*

Official SDKs

- Official **SDKs** (recommended): Python, TypeScript, Java
- *https://docs.anthropic.com/en/docs/get-started#install-the-sdk*

```
pip install anthropic
```

```python
import anthropic

client = anthropic.Anthropic()

message = client.messages.create(
    model="claude-opus-4-20250514",
    max_tokens=1000,
    temperature=1,
    system="You are a world-class poet. Respond only with short poems.",
    messages=[
        {
            "role": "user",
            "content": [
                {
                    "type": "text",
                    "text": "Why is the ocean salty?"
                }
            ]
        }
    ]
)
print(message.content)
```

Prompt Structure and Tuning

- **message roles** - a system message inserted into the model's context
- **temperature** - randomness
- max_tokens - limit response length
- messages - specify the prior conversational turns

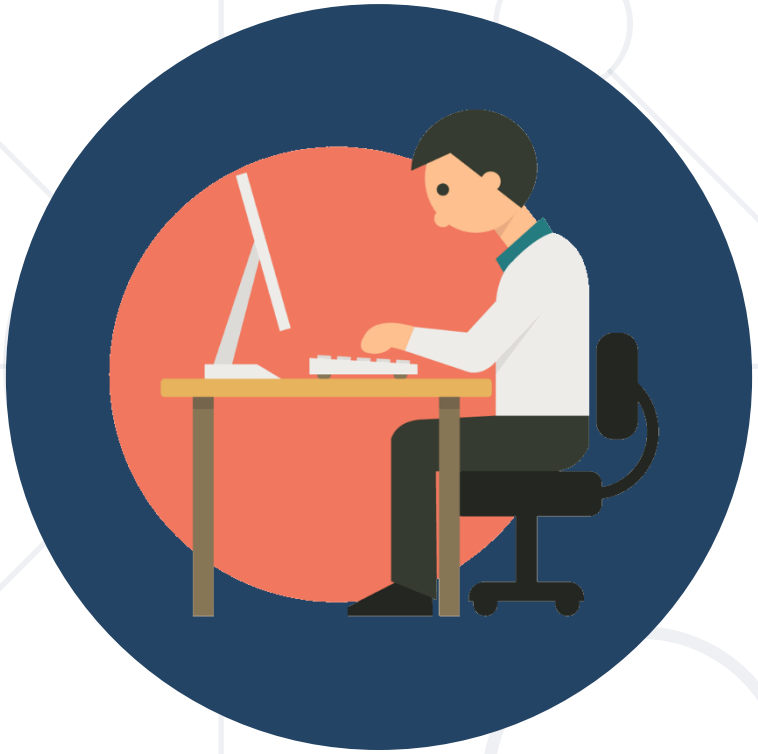
Working with Files

- Create a file

```
POST /v1/files
```

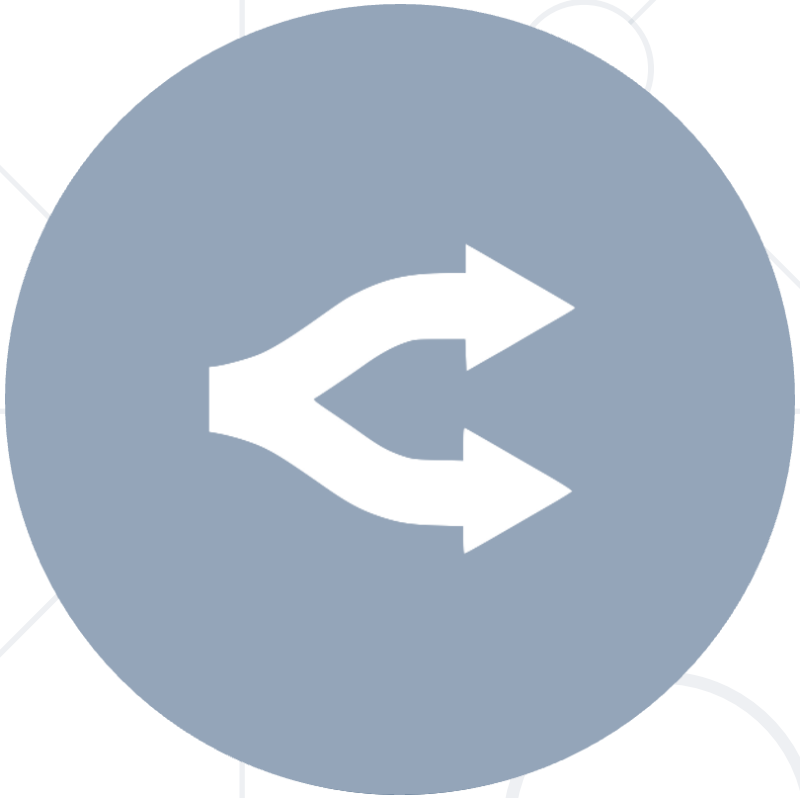
- Reference it in the messages

```
messages=[
    {
        "role": "user",
        "content": [
            { "type": "text", "text": "Classify this document:" },
            { "type": "document", "source": { "type": "file", "file_id": file_id } }
        ]
    }
],
```



Practice

Live Exercise – Setup Anthropic API Integration

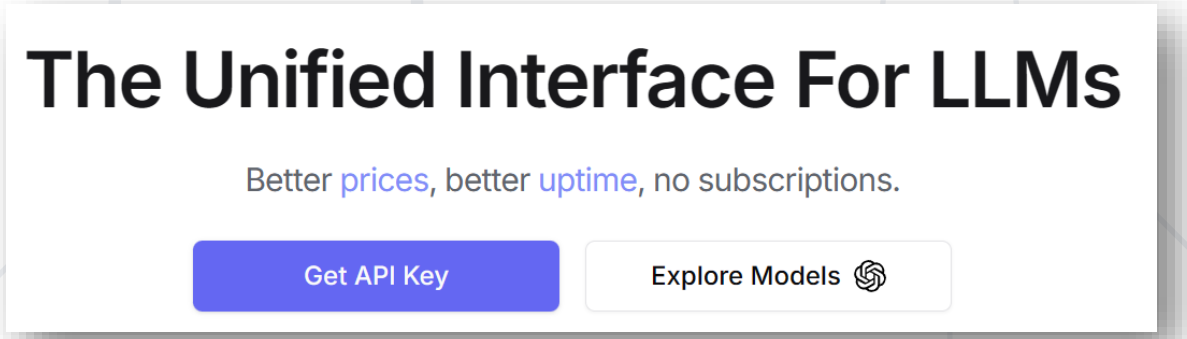


OpenRouter

One Interface to Rule Them All

OpenRouter – Universal Interface for LLMs

- **Unified API Aggregator:** A proxy layer that provides access to 200+ models through a single endpoint.
- **Protocol Standardization:** Implements the OpenAI as a standard, making it a "drop-in" replacement for existing OpenAI integrations.
- **Infrastructure as a Service:** No need for multiple vendor-specific SDKs and API keys.
- **Centralized Billing:** Single system for all providers.



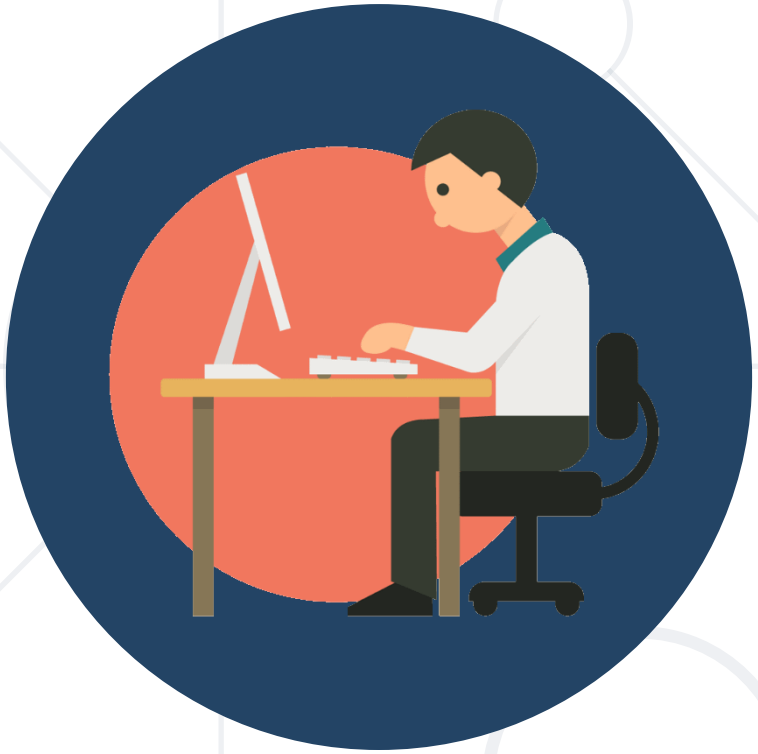
Why OpenRouter?

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- **Vendor Lock-in Mitigation:** Swap models by changing a single string in your config.
- **Automatic Fallbacks:** Configure "Router" chains that automatically switch to a backup provider if the primary one experiences downtime.
- **Performance Optimization:** Real-time monitoring of latency and throughput across different providers to route requests to the fastest instance.
- **Cost Efficiency:** Often cheaper than direct APIs due to bulk-reselling and access to specialized providers with aggressive pricing.

Developer Implementation

- **Base URL** Redirection
  - Change base_url to **https://openrouter.ai/api/v1**
  - Works **natively** with the official **OpenAI** libraries.
- **Open-Source Models:** Easy access to Llama 3, Qwen, and Mistral without hosting your own GPU clusters.
- **Advanced Headers:** Support for HTTP-Referer and X-Title for ranking and analytics in the OpenRouter dashboard.

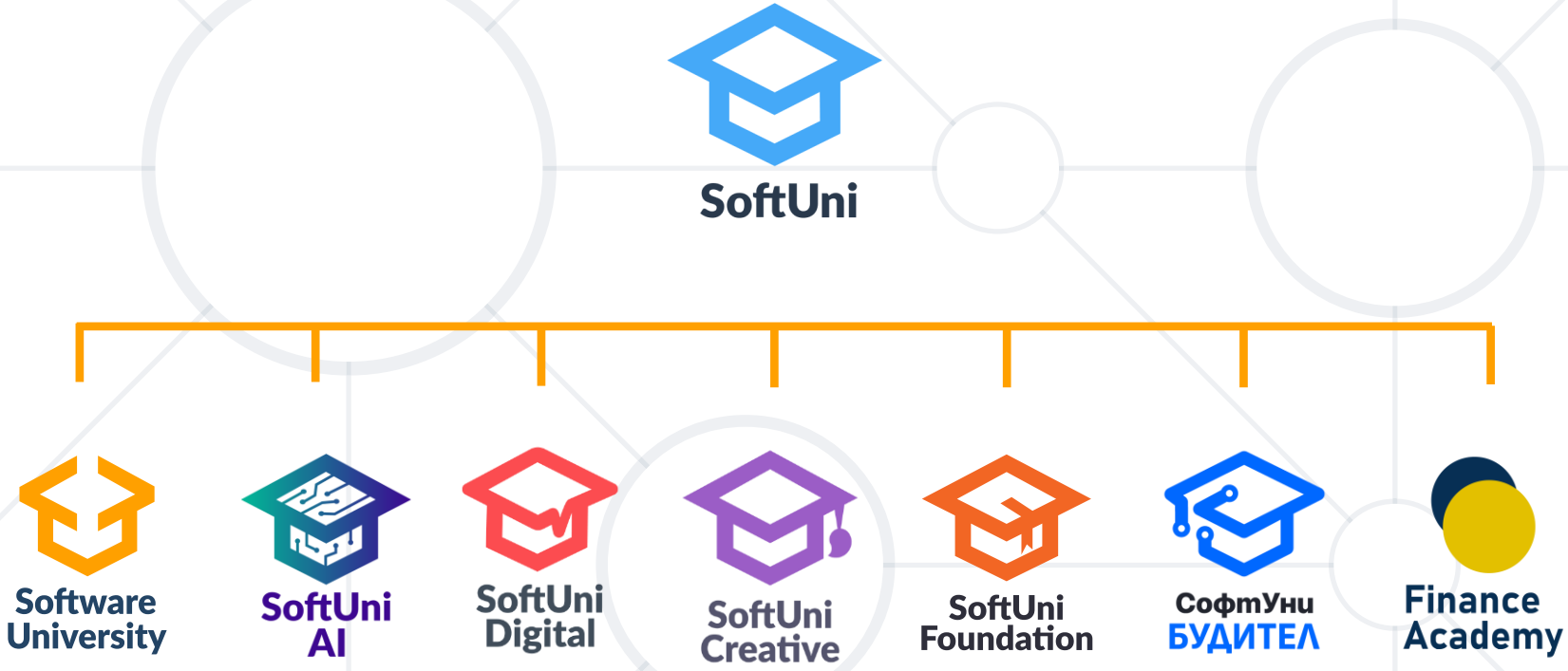


Practice

Live Exercise – Setup OpenRouter Integration

Questions?





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Summary

- OpenAI and Anthropic LLMs can be called via API
- APIs are rich and powerful, official SDKs are available
- OpenRouter is the universal interface for LLMs



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